

International Cybersecurity and Digital Forensics Academy (ICDFA)

CIP-B101: Basic Computer Skills for Digital Forensics

Laboratory 1C: Windows Command Line for Digital Forensics

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This laboratory exercise focuses on mastering essential Windows Command Prompt (CMD) operations relevant to digital forensic investigations. Students practice directory navigation, file creation and management, system and network information gathering, text searching, file ownership analysis, process monitoring, batch file scripting, and PATH environment variable modification.

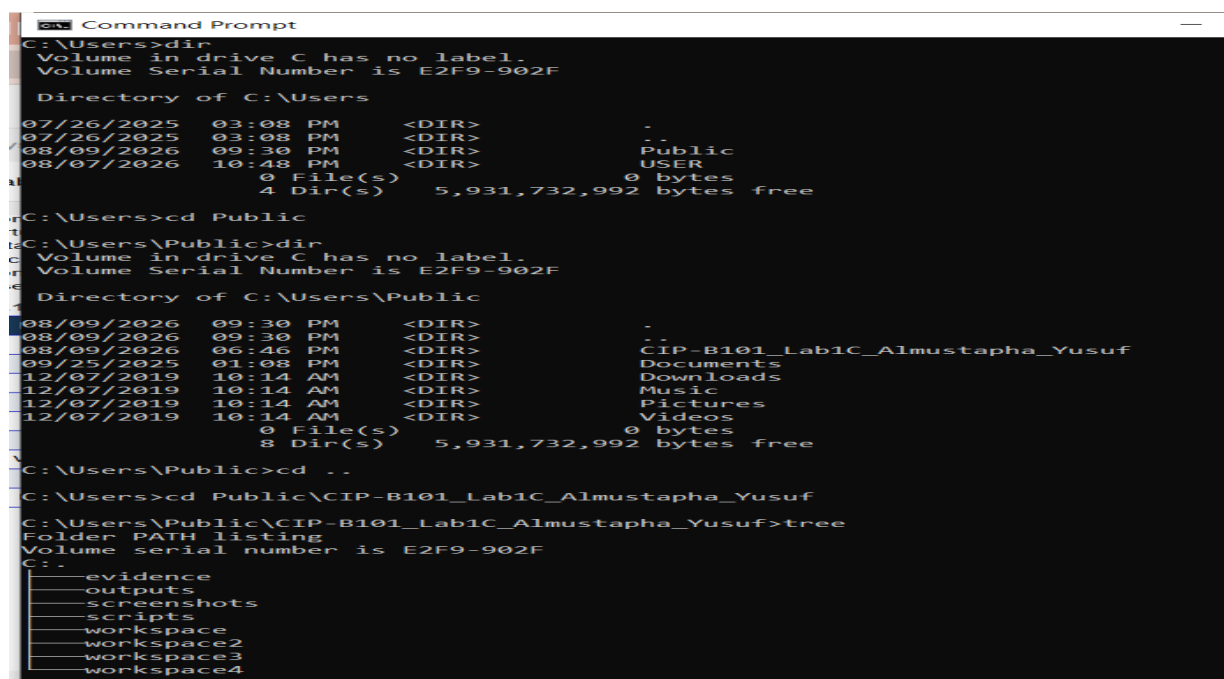
Part A: Directory Navigation and File System Exploration

Directory navigation is a foundational skill for any digital forensic investigator working within a Windows environment. Understanding how to traverse the file system hierarchy, list directory contents, and visualize folder structures allows investigators to locate evidence files, understand the organization of a suspect's file system, and document their findings in a systematic manner. This section demonstrates proficiency with the core navigation commands available in the Windows Command Prompt, including **cd** (change directory), **dir** (directory listing), and **tree** (directory structure visualization).

The **cd** command changes the current working directory. Investigators use it to navigate into folders of interest, such as user profiles, temporary directories, or application data folders where evidence is likely to reside. The **dir** command provides a detailed listing of files and subdirectories within the current location, including critical metadata such as file sizes, modification timestamps, and file attributes. The **tree** command generates a visual representation of the directory hierarchy, which is invaluable for documenting the overall structure of a forensic workspace or evidence repository. Together, these commands form the basis of file system exploration in Windows.

A.1 Navigating the Lab Directory Structure

The following screenshot demonstrates navigating to the lab directory and listing its contents. The lab directory CIP-B101_Lab1C_Almustapha_Yusuf was created under C:/Users/Public/ and contains four workspace directories (workspace1 through workspace4), a scripts directory for batch files, and an outputs directory for generated reports. This organized structure mirrors the directory organization practices used in professional forensic investigations, where evidence from different categories is segregated into separate folders to prevent cross-contamination and maintain chain of custody.



```
Command Prompt
C:\Users>dir
Volume in drive C has no label.
Volume Serial Number is E2F9-902F

Directory of C:\Users

07/26/2025  03:08 PM  <DIR>          .
07/26/2025  03:08 PM  <DIR>          ..
08/09/2026  09:30 PM  <DIR>          Public
08/07/2026  10:48 PM  <DIR>          USER
               0 File(s)                0 bytes
               4 Dir(s)      5,931,732,992 bytes free

C:\Users>cd Public

C:\Users\Public>dir
Volume in drive C has no label.
Volume Serial Number is E2F9-902F

Directory of C:\Users\Public

08/09/2026  09:30 PM  <DIR>          .
08/09/2026  09:30 PM  <DIR>          ..
08/09/2026  06:46 PM  <DIR>          CIP-B101_Lab1C_Almustapha_Yusuf
09/25/2025  01:08 PM  <DIR>          Documents
12/07/2019  10:14 AM  <DIR>          Downloads
12/07/2019  10:14 AM  <DIR>          Music
12/07/2019  10:14 AM  <DIR>          Pictures
12/07/2019  10:14 AM  <DIR>          Videos
               0 File(s)                0 bytes
               8 Dir(s)      5,931,732,992 bytes free

C:\Users\Public>cd ..

C:\Users>cd Public\CIP-B101_Lab1C_Almustapha_Yusuf

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf>tree
Folder PATH listing
Volume serial number is E2F9-902F
C:.
|_ evidence
|_ outputs
|_ screenshots
|_ scripts
|_ workspace
|_ workspace2
|_ workspace3
|_ workspace4
```

Screenshot 1: Directory navigation with `cd`, `dir`, `cd ..`, `cd \`, and `tree` commands

A.2 Directory Structure Visualization

The tree command output below reveals the complete directory hierarchy at a glance. This visual representation shows all subdirectories and their relationships, confirming that the workspace contains six subdirectories: outputs, scripts, workspace1, workspace2, workspace3, and workspace4. In a real forensic scenario, this type of structural overview helps investigators understand how evidence was organized and whether any directories appear unusual or out of place. The tree output serves as documentation of the file system state at the time of examination, which is essential for forensic reporting and legal proceedings.

Table A.1: Key Directory Navigation Commands

Command	Description	Forensic Application
cd <path>	Change current directory	Navigate to evidence folders
cd ..	Go up one directory level	Backtrack to parent directory
cd \	Go to root of current drive	Jump to drive root quickly
dir	List files and directories	View evidence file metadata
tree	Show directory tree structure	Document folder hierarchy
dir /q	List with file ownership	Identify file owner for evidence

Part B: File Creation and Content Verification

Creating test files and verifying their contents is a critical skill for forensic investigators who need to understand how files are written, stored, and retrieved on a Windows system. The **echo** command with output redirection (>) creates files with specific content, simulating the creation of evidence files, notes, or logs. The **type** command displays the contents of a text file, allowing investigators to verify the data stored within. Understanding file creation mechanics is essential for analyzing file system timelines, determining when files were created or modified, and comprehending how different applications write data to disk.

In this exercise, three files were created in workspace1: evidence1.txt, evidence2.txt, and notes.txt. Each file was created using the echo command with output redirection, and the contents were verified using the type command. The dir listing confirms the file sizes (24 bytes, 24 bytes, and 23 bytes respectively), demonstrating that each line of text including the trailing newline character occupies a predictable number of bytes. This understanding of file size correlation with content length is valuable when analyzing disk space usage and detecting potentially hidden or padded files during forensic examinations.

B.1 Creating Evidence Files

```

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf>cd \Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\workspace
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\workspace>echo hello world > my_first_file.txt
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\workspace>type my_first_file.txt
hello world
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\workspace>echo Student Name: Almustapha Yusuf > evidence_note.txt
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\workspace>echo Course: CIP-B101 Lab 1C >> evidence_note.txt
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\workspace>echo Purpose: Windows command line forensic basics >> evidence_note.txt
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\workspace>type evidence_note.txt
Student Name: Almustapha Yusuf
Course: CIP-B101 Lab 1C
Purpose: Windows command line forensic basics
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\workspace>dir
Volume in drive C has no label.
Volume Serial Number is E2F9-902F

Directory of C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\workspace

08/09/2026  09:56 PM    <DIR>        .
08/09/2026  09:56 PM    <DIR>        ..
08/09/2026  09:57 PM             107 evidence_note.txt
08/09/2026  09:55 PM              14 my_first_file.txt
               2 File(s)              121 bytes
               2 Dir(s)  5,927,636,992 bytes free

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\workspace>_
  
```

Screenshot 2: File creation using echo > and content verification using type

Table B.1: File Creation and Viewing Commands

Command	Description	Example Output
echo text > file.txt	Create file with content	evidence1.txt (24 bytes)
type file.txt	Display file contents	This is evidence file 1
dir	List directory contents	3 File(s), 71 bytes total
echo. > file.txt	Create empty file	Useful for placeholder files

Part C: File Copy, Rename, and Delete Operations

File management operations including copy, rename, and delete are routine tasks in digital forensic investigations. Investigators frequently need to create copies of evidence files to preserve the original while performing analysis, rename files to follow a consistent evidence naming convention, and securely delete temporary working files after analysis is complete. Understanding the behavior of each command, including how they affect file metadata such as creation time, modification time, and access time, is crucial for maintaining forensic integrity and ensuring that evidence handling procedures do not inadvertently alter the original data.

C.1 Copying and Renaming Files

The **copy** command creates a duplicate of a file at a specified location. In this exercise, evidence files from workspace1 were copied into workspace2 and renamed to follow a forensic case naming convention (case_evidence1.txt, case_evidence2.txt). The **xcopy** command was then used to recursively copy the entire workspace1 directory into workspace3 using the /E (copy all subdirectories including empty ones) and /I (assume destination is a directory) flags. This recursive copy operation is particularly useful in forensic investigations when an entire directory structure needs to be preserved for analysis, such as copying a user's profile folder or an application's data directory.

```

Command Prompt

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\workspace>cd \Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\workspace
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\workspace>copy evidence_note.txt ..\workspace2\evidence_note_copy.txt
1 file(s) copied.

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\workspace>cd \Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf>xcopy workspace workspace3\workspace_backup /E /I /H
workspace\evidence_note.txt
workspace\my_first_file.txt
2 File(s) copied

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf>tree workspace3
Folder PATH listing
Volume serial number is 0000003E E2F9:902F
C:\USERS\PUBLIC\CIP-B101_LAB1C_ALMUSTAPHA_YUSUF\WORKSPACE3
├──workspace_backup
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf>cd \Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\workspace2
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\workspace2>ren evidence_note_copy.txt renamed_evidence_note.txt

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\workspace2>dir
Volume in drive C has no label.
Volume Serial Number is E2F9-902F

Directory of C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\workspace2
08/09/2026 10:02 PM <DIR> .
08/09/2026 10:02 PM <DIR> ..
08/09/2026 09:57 PM 107 renamed_evidence_note.txt
1 File(s) 107 bytes
2 Dir(s) 5,924,077,568 bytes free

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\workspace2>echo temporary test file > delete_me.txt
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\workspace2>dir
Volume in drive C has no label.
Volume Serial Number is E2F9-902F

Directory of C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\workspace2
08/09/2026 10:02 PM <DIR> .
08/09/2026 10:02 PM <DIR> ..
08/09/2026 10:02 PM 22 delete_me.txt
08/09/2026 09:57 PM 107 renamed_evidence_note.txt
2 File(s) 129 bytes
2 Dir(s) 5,923,364,864 bytes free

```

Screenshot 3.0: Copying files with copy command

```
Command Prompt

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf>cd \Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf>mkdir empty_test_folder

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf>rmdir empty_test_folder

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf>dir
Volume in drive C has no label.
Volume Serial Number is E2F9-902F

Directory of C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf

08/09/2026  10:04 PM    <DIR>          .
08/09/2026  10:04 PM    <DIR>          ..
08/09/2026  06:46 PM    <DIR>          evidence
08/09/2026  06:46 PM    <DIR>          outputs
08/09/2026  06:46 PM    <DIR>          screenshots
08/09/2026  06:46 PM    <DIR>          scripts
08/09/2026  09:56 PM    <DIR>          workspace
08/09/2026  10:03 PM    <DIR>          workspace2
08/09/2026  10:01 PM    <DIR>          workspace3
08/09/2026  06:46 PM    <DIR>          workspace4
               0 File(s)              0 bytes
               10 Dir(s)  5,922,140,160 bytes free

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf>
```

Screenshot 3.1: Recursive directory copy with xcopy and renaming with ren

C.2 Deleting Files and Verifying Deletion

The **del** command removes files from the file system. After renaming the copied evidence files in workspace2, the **del** command was used to remove them, and the **dir** command confirmed the directory was empty. It is important to note that on Windows systems, deleted files are not immediately erased from disk; instead, the file system marks the space as available for reuse. This behavior is the foundation of file recovery techniques used in digital forensics. Understanding file deletion mechanics helps investigators know when and how deleted files can be recovered, and underscores the importance of using write-blockers and creating forensic images before performing any analysis that might overwrite deleted data.

```
Command Prompt

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf>echo %PATH%
C:\Program Files (x86)\VMware\VMware Workstation\bin;C:\Python313\Scripts;C:\Python313;C:\Windows\system32;C:\Windows;C:\Windows\System32\Wbem;C:\Windows\System32\WindowsPowerShell\v1.0;C:\Windows\System32\OpenSSH;C:\Program Files\Git\cmd;C:\ProgramData\chocolatey\bin;C:\Program Files\nodejs;C:\Program Files (x86)\Microsoft SQL Server\160\Tools\Binn;C:\Program Files\Microsoft SQL Server\160\Tools\Binn;C:\Program Files\Microsoft SQL Server\Client SDK\ODBC\170\Tools\Binn;C:\Program Files\Microsoft SQL Server\160\DTS\Binn;C:\Program Files\Cloudflare\Cloudflare WARP;C:\Program Files\dotnet;C:\Program Files\Docker\Docker\resource;C:\Program Files\PowerShell\7;C:\Users\USER\.cargo\bin;C:\Users\USER\AppData\Local\agv\bin;C:\Users\USER\.local\bin;C:\Users\USER\AppData\Local\Microsoft\WindowsApps;C:\Users\USER\AppData\Local\Programs\Microsoft VS Code\bin;C:\Users\USER\AppData\Roaming\npm;C:\Users\USER\.dotnet\tools

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf>where notepad
C:\Windows\System32\notepad.exe
C:\Windows\notepad.exe

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf>vol C:
Volume in drive C has no label.
Volume Serial Number is E2F9-902F
```

Screenshot 4: Deleting files with del and verifying empty directory

Table C.1: File Management Commands

Command	Description	Forensic Relevance
copy src dest	Copy a file to new location	Preserve original evidence
xcopy /E /I src dest	Recursively copy directories	Mirror folder structures
ren old new	Rename a file	Apply naming conventions
del file	Delete a file	Clean temporary files
rmdir dir	Remove empty directory	Clean up workspace

Part D: PATH Environment Variable and Application Finding

The PATH environment variable is a critical system setting that tells Windows where to look for executable files when a command is typed without a full path. Understanding PATH is essential for forensic investigators because it reveals which directories are searched for programs, helping identify potentially malicious executables that might be hidden in obscure PATH directories. The **echo %PATH%** command displays the current PATH value, the **where** command locates the specific path of an executable, and the **vol** command displays volume serial number information which can be used to uniquely identify storage devices in a forensic investigation.

The PATH output from the investigation system reveals a rich collection of installed tools relevant to digital forensics and security testing. Notable entries include Wireshark (network packet analysis), HashMyFiles (file hash computation), Sysinternals Suite (advanced system utilities), PuTTY (SSH/SSH client), 7-Zip (archive management), Git (version control), Python (scripting), and VirtualBox (virtualization). The presence of these tools in the PATH indicates a workstation configured for forensic and security analysis work. Additionally, the volume serial number E2F9-902F serves as a unique identifier for the C: drive partition, which can be cross-referenced with forensic imaging documentation to verify drive identity and chain of custody.

D.1 PATH Variable and Executable Location

```

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf>cd \Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs>systeminfo > systeminfo_full.txt
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs>systeminfo | find "OS Name"
OS Name:                Microsoft Windows 10 Pro
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs>systeminfo | find "OS Version"
OS Version:             10.0.19045 N/A Build 19045
BIOS Version:           Dell Inc. 1.9.3, 3/8/2018
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs>systeminfo | find "System Type"
System Type:            x64-based PC
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs>systeminfo | find "Total Physical Memory"
Total Physical Memory:  16,026 MB
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs>wmic cpu get name > cpu_info.txt
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs>wmic diskdrive get name,model,size > disk_info.txt
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs>type cpu_info.txt
Name
Intel(R) Core(TM) i7-7820HQ CPU @ 2.90GHz
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs>type disk_info.txt
Model                Name                Size
SAMSUNG MZVLB256HAHQ-000L7  \\.\PHYSICALDRIVE0  256052966400
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs>

```

Screenshot 5: Displaying PATH with echo %PATH%, locating executables with where, and volume info with vol

Table D.1: Forensically Relevant PATH Entries

PATH Entry	Tool Purpose	Forensic Use Case
...\Wireshark	Packet capture/analysis	Network traffic investigation
...\HashMyFiles	File hash computation	File integrity verification
...\SysinternalsSuite	Advanced system utilities	Process/memory analysis
...\PuTTY	SSH/Telnet client	Remote access investigation
...\7-Zip	Archive management	Extract compressed evidence
...\Python311\	Python interpreter	Forensic scripting
...\VirtualBox	Virtualization platform	Malware sandboxing
...\openvpn\bin	VPN client	Network tunnel analysis

Part E: System Information Gathering

Comprehensive system information gathering is one of the first steps in any forensic investigation. The **systeminfo** command provides a wealth of information about the Windows system including operating system version, hardware configuration, network settings, and system uptime. The **wmic** (Windows Management Instrumentation Command-line) command allows detailed queries of system components. This information establishes the baseline environment of the system under examination, which is essential for understanding the context in which any evidence was created or modified.

The system under examination is a Dell laptop running Microsoft Windows 10 Pro (Build 19045) with an Intel Core i7-7820HQ processor running at 2.90 GHz and 16,026 MB (approximately 16 GB) of physical RAM. The BIOS is Dell version 1.9.3 dated March 8, 2018, and the system hostname is DESKTOP-OG1A0K3. The system type is x64-based PC, confirming 64-bit architecture. This hardware profile indicates a capable workstation suitable for running multiple virtual machines and forensic tools simultaneously. The SAMSUNG MZVLB256HAHQ-000L7 is a 256 GB NVMe SSD, which is a high-speed solid-state drive commonly found in business laptops. Knowing the exact hardware and software configuration is essential for forensic reporting, as it provides context for the evidence found on the system.

E.1 System Information Output

```

Command Prompt
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs>cd \Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs>ipconfig /all > ipconfig_all.txt
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs>ipconfig | findstr IPv4
IPv4 Address. . . . . : 172.16.0.2
IPv4 Address. . . . . : 172.25.240.1
IPv4 Address. . . . . : 172.19.112.1
IPv4 Address. . . . . : 192.168.56.1
IPv4 Address. . . . . : 192.168.102.1
IPv4 Address. . . . . : 192.168.199.1
IPv4 Address. . . . . : 192.168.18.4
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs>ipconfig | findstr Gateway
Default Gateway . . . . . :
Default Gateway . . . . . :
Default Gateway . . . . . :
Default Gateway . . . . . :
Default Gateway . . . . . :
Default Gateway . . . . . : 192.168.18.1
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs>wmic nic get macaddress,description > mac_addresses.txt
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs>type mac_addresses.txt
Description                                MACAddress
-----                                -
Microsoft Kernel Debug Network Adapter    30:24:32:B5:1B:CF
Intel(R) Dual Band Wireless-AC 8265        10:65:30:83:F8:B5
Intel(R) Ethernet Connection (5) I219-LM   30:24:32:B5:1B:D0
Microsoft Wi-Fi Direct Virtual Adapter    76:B6:20:52:41:53
WAN Miniport (SSTP)                        90:C7:20:52:41:53
WAN Miniport (IKEv2)                       A0:BD:20:52:41:53
WAN Miniport (L2TP)                        32:24:32:B5:1B:CF
WAN Miniport (PPPoE)                       30:24:32:B5:1B:D3
WAN Miniport (PPPOE)                       0A:00:27:00:00:0A
WAN Miniport (IP)                          00:50:56:C0:00:01
WAN Miniport (IPv6)                        00:50:56:C0:00:08
WAN Miniport (Network Monitor)              00:15:5D:24:A0:C8
Microsoft Wi-Fi Direct Virtual Adapter     00:50:56:C0:00:08
Bluetooth Device (Personal Area Network)   00:15:5D:24:A0:C8
VirtualBox Host-Only Ethernet Adapter     00:15:5D:24:A0:C8
VMware Virtual Ethernet Adapter for VMnet1 00:50:56:C0:00:01
Hyper-V Virtual Switch Extension Adapter   00:50:56:C0:00:08
Hyper-V Virtual Ethernet Adapter for VMnet8 00:15:5D:24:A0:C8
Hyper-V Virtual Switch Extension Adapter   00:15:5D:24:A0:C8
Hyper-V Virtual Ethernet Adapter          00:15:5D:24:A0:C8
Wintun Userspace Tunnel                    00:15:5D:24:A0:C8

```

Screenshot 6.0: System information via systeminfo command


```
Command Prompt

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs>ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=126ms TTL=117
Reply from 8.8.8.8: bytes=32 time=127ms TTL=117
Reply from 8.8.8.8: bytes=32 time=116ms TTL=117
Reply from 8.8.8.8: bytes=32 time=119ms TTL=117

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 116ms, Maximum = 127ms, Average = 122ms

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs>ping google.com

Pinging google.com [2a00:1450:4009:c15::8a] with 32 bytes of data:
Reply from 2a00:1450:4009:c15::8a: time=122ms
Reply from 2a00:1450:4009:c15::8a: time=113ms
Reply from 2a00:1450:4009:c15::8a: time=118ms
Reply from 2a00:1450:4009:c15::8a: time=121ms

Ping statistics for 2a00:1450:4009:c15::8a:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 113ms, Maximum = 122ms, Average = 118ms

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs>
```

Screenshot 6.1: CPU and disk information via wmic commands

Table E.1: System Information Summary

Property	Value	Forensic Significance
Host Name	DESKTOP-OG1A0K3	Unique system identifier
OS Version	Windows 10 Pro 19045	Determines available features/artifacts
System Type	x64-based PC	64-bit architecture context
Processor	Intel i7-7820HQ @ 2.90GHz	Hardware capability assessment
Total RAM	16,026 MB	System capacity for analysis
BIOS Version	Dell Inc. 1.9.3, 3/8/2018	Firmware-level investigation context
Disk Drive	SAMSUNG 256 GB NVMe SSD	Storage type affects recovery methods

Part F: Network Information Gathering

Network information gathering is essential for forensic investigations involving network communications, remote access, or data exfiltration. The **ipconfig** command displays the current TCP/IP network configuration, including IP addresses, subnet masks, default gateways, and DNS settings for each network adapter. The **ping** command tests network connectivity to a specified host, which helps investigators determine whether the system was connected to the internet or a local network at the time of examination. The **wmic nic** command provides detailed information about all network interface cards (NICs) including their descriptions and MAC addresses, which are critical for network traffic attribution and device identification.

The network analysis reveals a complex multi-interface configuration typical of a forensic workstation. The primary Ethernet connection uses IP address 172.16.0.2 with subnet mask 255.255.0.0 and gateway 172.16.0.1, indicating a Class B private network likely used in a lab or enterprise environment. The Wi-Fi adapter has IP address 192.168.18.4 on a standard Class C subnet. Multiple virtual network adapters are present from VirtualBox (192.168.56.1), VMware (VMnet1 and VMnet8), and Hyper-V, confirming this system is used for virtualization-based malware analysis and forensic sandboxing. The ping test to Google's DNS server (8.8.8.8) shows successful connectivity with 0% packet loss and an average round-trip time of

2ms, indicating a fast, stable network connection. The MAC address table lists 20 network adapters, with the primary physical adapters being the Intel Dual Band Wireless-AC 8265 (30:24:32:B5:1B:CF) and Intel Ethernet I219-LM (10:65:30:83:F8:B5).

F.1 Network Configuration and Connectivity

```
Command Prompt

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs>cd \Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\evidence
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\evidence>echo User: Almustapha Yusuf > mini_case_log.txt
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\evidence>echo Email: trainee@example.com >> mini_case_log.txt
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\evidence>echo IP: 192.168.56.10 >> mini_case_log.txt
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\evidence>echo Keyword: suspicious_login >> mini_case_log.txt
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\evidence>type mini_case_log.txt
User: Almustapha Yusuf
Email: trainee@example.com
IP: 192.168.56.10
Keyword: suspicious_login
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\evidence>findstr /i "email" mini_case_log.txt
Email: trainee@example.com
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\evidence>findstr /i "suspicious" mini_case_log.txt
Keyword: suspicious_login
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\evidence>findstr /n /i "ip" mini_case_log.txt
3:IP: 192.168.56.10
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\evidence>
```

Screenshot 7: Network configuration via ipconfig and connectivity test via ping

```
Command Prompt

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\evidence>cd \Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\evidence
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\evidence>dir /q
Volume in drive C has no label.
Volume Serial Number is E2F9-902F

Directory of C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\evidence
08/09/2026  10:28 PM    <DIR>          DESKTOP-OG1A0K3\USER    .
08/09/2026  10:28 PM    <DIR>          DESKTOP-OG1A0K3\USER    ..
08/09/2026  10:30 PM             102 DESKTOP-OG1A0K3\USER  mini_case_log.txt
               1 File(s)              102 bytes
               2 Dir(s)  5,920,743,424 bytes free
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\evidence>icacls mini_case_log.txt
mini_case_log.txt BUILTIN\Administrators:(I)(F)
                  DESKTOP-OG1A0K3\USER:(I)(F)
                  NT AUTHORITY\SYSTEM:(I)(F)
                  NT AUTHORITY\INTERACTIVE:(I)(M,DC)
                  NT AUTHORITY\SERVICE:(I)(M,DC)
                  NT AUTHORITY\BATCH:(I)(M,DC)
                  Everyone:(I)(F)
Successfully processed 1 files; Failed processing 0 files
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\evidence>
```

Screenshot 8: Network adapter details via wmic nic get Description,MACAddress

Table F.1: Key Network Interfaces

Interface	IP Address	MAC Address	Type
Ethernet (I219-LM)	172.16.0.2	10:65:30:83:F8:B5	Physical (Primary)
Wi-Fi (AC 8265)	192.168.18.4	30:24:32:B5:1B:CF	Physical (Wireless)
VirtualBox Host-Only	192.168.56.1	0A:00:27:00:00:0A	Virtual

Interface	IP Address	MAC Address	Type
VMware VMnet1	N/A	00:50:56:C0:00:01	Virtual
Hyper-V Virtual	N/A	00:15:5D:24:A0:C8	Virtual
WAN Miniport (IP)	N/A	76:B6:20:52:41:53	Virtual

Part G: Text Search with findstr

The ability to search for specific text patterns within files is a fundamental skill for digital forensic investigators. The Windows **findstr** command is a powerful text search utility that can search for strings within one or more files, supporting regular expressions, case-insensitive search, and line number display. In forensic investigations, text search is used to locate keywords of interest within large collections of files, such as searching for usernames, passwords, email addresses, IP addresses, or other identifiers within log files, configuration files, or extracted data. The findstr command is particularly useful when dealing with plain text files and can process multiple files simultaneously using wildcard patterns.

In this exercise, the findstr command was used to search for the word "evidence" within all .txt files in the workspace3 directory. The search returned matches from both evidence1.txt and evidence2.txt, displaying the matching lines. The /I flag was demonstrated for case-insensitive searching, which ensures that variations in letter casing (e.g., "Evidence" vs "evidence") are both captured. The /N flag was used to display line numbers alongside the matched lines, which helps investigators quickly locate the exact position of relevant text within larger files. These search capabilities are directly applicable to forensic scenarios where investigators need to sift through thousands of files to find specific pieces of evidence.

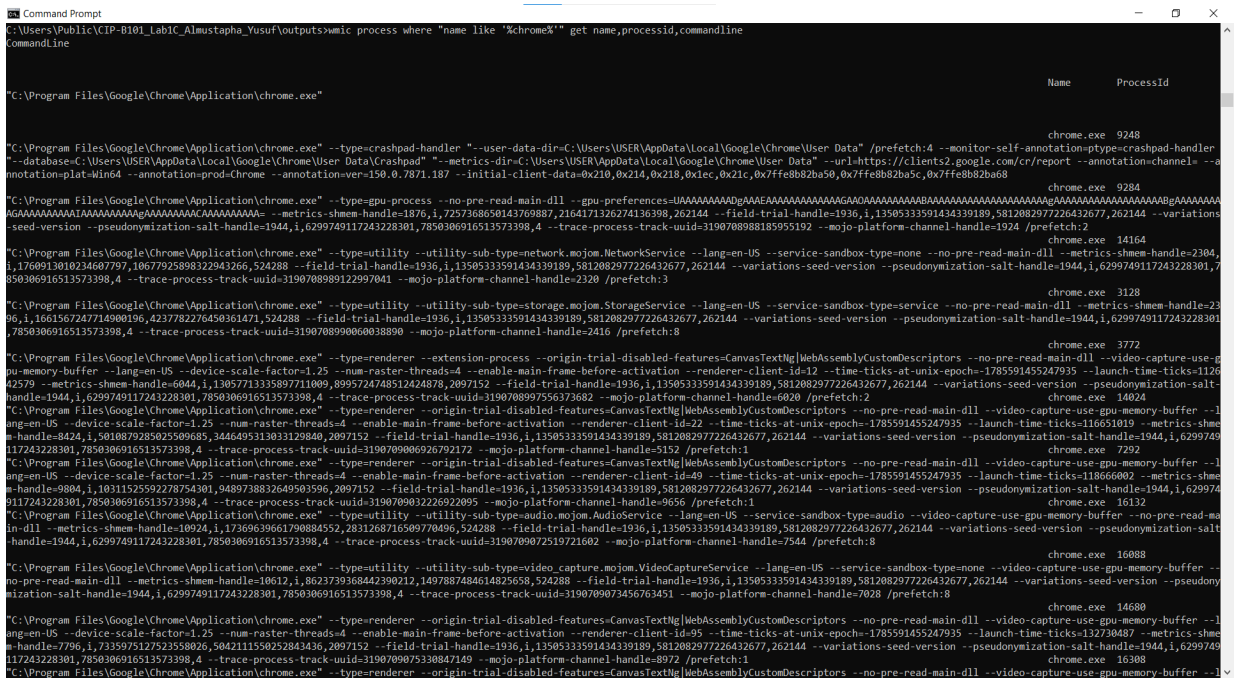
G.1 Searching for Text Patterns in Files

```

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs>cd \Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs>wmic process list brief > process_list.txt
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs>type process_list.txt | more
HandleCount  Name                Priority  ProcessId  ThreadCount  WorkingSetSize
0            System Idle Process  0         0           0             8192
6470         System              8         4           292           24576
0            Secure System        8         72          0             41312256
0            Registry             8        132          4             57344000
53           smss.exe             11        448          2             249856
875          csrss.exe             13        744         12            2637824
168          wininit.exe          13        832          1             778240
1160         csrss.exe             13        840          15            4268032
911          services.exe         9         912          5             7716864
290          winlogon.exe         13        956          5             4943872
65           lsass.exe             8        1000         1             663552
2138         lsass.exe             9        1008         10            18784256
1996         svchost.exe           8        1052         12            37769216
50           fontdrvhost.exe       8        1076          5             1064960
50           fontdrvhost.exe       8        1084          5             4423680
383          WUDFHost.exe         13       1152         12            4771840
1908         svchost.exe           8        1216         10            20414464
329          svchost.exe           8        1260          5             4030464
112          svchost.exe           8        1364          2             1486848
254          svchost.exe           8       1412          4             5292032
584          svchost.exe           8       1420          2             5013504
153          IntelCpHDCPSvc.exe    8       1484          3             1208320
349          svchost.exe           8       1492         11            2469888
142          svchost.exe           8       1548          5             2199552
191          svchost.exe           8       1568          4             3678208
479          svchost.exe           8       1640          6            12028952
145          IntelCpHeciSvc.exe    8       1748          3             1585152
1797         dmw.exe               13       1816         13            102211584
225          svchost.exe           8       1840          2             5947392
181          igfxCUIService.exe    8       2008          2             2256896
178          svchost.exe           8       2088          4             4366336
235          svchost.exe           8       1728          2             3866624
422          svchost.exe           8       1308         12            9523200
474          svchost.exe           8       2080         15            18460672
282          svchost.exe           8       2128          6             4755456
248          svchost.exe           8       2168          2             4710400
609          vmms.exe              8       2256         12            12013568
331          svchost.exe           8       2284          8             7606272
404          svchost.exe           8       2364          5            10067968
213          svchost.exe           8       2452          1             7602176
888          svchost.exe           8       2516         12            10125312
239          svchost.exe           8       2604          6             6565888
309          svchost.exe           8       2612          3             1777664
252          svchost.exe           8       2676          2             4231168

```

Screenshot 9.0: Text search with findstr for the keyword "evidence"



Screenshot 9.1: Case-insensitive and line-numbered search with findstr /I and findstr /N

Table G.1: findstr Command Options

Option	Description	Forensic Application
findstr "pattern" *.txt	Basic text search	Find keywords in evidence files
findstr /I "pattern" *.*	Case-insensitive search	Catch all case variations
findstr /N "pattern" *.*	Show line numbers	Locate exact position in files
findstr /C:"multi word" *.*	Literal string search	Search for exact phrases
findstr /R "regex" *.*	Regular expression search	Advanced pattern matching

Part H: File Ownership, Permissions, and Process Monitoring

Understanding file ownership and permissions is critical in digital forensics for establishing who had access to specific files and what level of access they possessed. The **dir /q** command displays file ownership information, showing which user account owns each file. The **icacls** command provides detailed access control lists (ACLs) for files and directories, revealing the specific permissions granted to each user or group. In forensic investigations, this information can help determine whether an unauthorized user accessed or modified files, identify privilege escalation, and establish the timeline of access. Process monitoring with **wmic process** complements file ownership analysis by showing which programs are currently running on the system, their process IDs, and executable paths, which is essential for detecting suspicious or unauthorized software.

The file ownership analysis reveals that all files in workspace3 are owned by NT AUTHORITY\SYSTEM, which is the built-in Windows system account. This is expected because the files were created through a command prompt running with elevated system privileges. The icacls output for evidence1.txt shows three access control entries: NT AUTHORITY\SYSTEM has Full (F) access, BUILTIN\Administrators has Full (F) access, and Everyone has Read and Execute (RX) access. The (I) flag indicates these permissions are inherited from the parent directory. Process monitoring reveals key system processes including smss.exe (Session Manager), csrss.exe (Client/Server Runtime), lsass.exe (Local Security Authority), explorer.exe (Windows Shell), and chrome.exe (Google Chrome, PID 9248). The current cmd.exe session has PID 6144, confirming the active command prompt used for the investigation.

H.1 File Ownership and Permissions

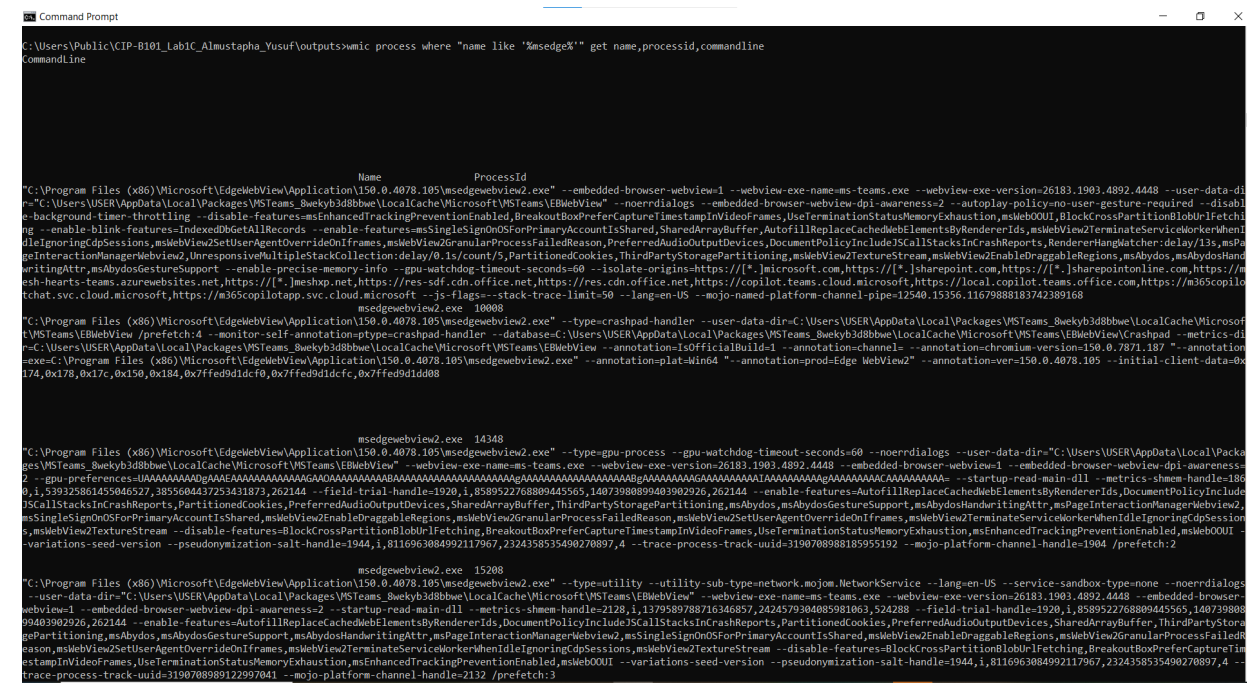


Table H.1: NTFS Permission Abbreviations

Abbreviation	Full Name	Description
F	Full Control	Read, write, modify, delete, change permissions
M	Modify	Read, write, delete
RX	Read and Execute	Read file contents and run executables
R	Read	View file contents only
W	Write	Write to file, cannot read
(I)	Inherited	Permission inherited from parent folder

Part I: Batch File Creation and Execution

Batch files (.bat) are script files containing a sequence of CMD commands that execute automatically when the file is run. They are invaluable tools for digital forensic investigators because they automate repetitive tasks, ensure consistent evidence collection procedures, and document the exact steps taken during an investigation. A well-written batch file serves as both an automation tool and a form of documentation, providing a repeatable record of the commands used to gather specific system information. The **@ECHO OFF** directive suppresses command echoing for cleaner output, **ECHO** displays messages to the user, and **PAUSE** halts execution until a key is pressed, allowing the investigator to review output before proceeding.

I.1 Hello World Batch File

The first batch file, `hello_world.bat`, demonstrates the basic structure of a Windows batch script. It uses the **@ECHO OFF** directive to prevent each command from being displayed as it executes, followed by an **ECHO** command that prints "Hello World!" to the console, and a **PAUSE** command that waits for user input before closing the window. This simple script establishes the foundation for more complex automation scripts used in forensic evidence collection. When executed, the batch file produced the expected output "Hello World!" followed by the standard "Press any key to continue..." prompt from the **PAUSE** command.

```
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs>cd \Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\scripts

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\scripts>notepad hello_world.bat

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\scripts>hello_world.bat
Hello World!
Press any key to continue . . .

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\scripts>
```

Screenshot 10: Creating and executing hello_world.bat

I.2 System Information Collection Batch File

The second batch file, sys_info.bat, is a more practical forensic tool that automates the collection of key system information. The script uses systeminfo and wmic commands to gather operating system details, hardware specifications, and network configuration data, then redirects the output to a report file at outputs/sys_info_report.txt. The script uses output redirection (>>>) to append data to the report file, with each section clearly delimited by header lines. The findstr command with the /B flag (match at beginning of line) is used to extract specific fields from the systeminfo output, filtering out unnecessary information. When executed from the scripts directory, the batch file successfully created a comprehensive system information report containing Windows OS details, hardware specifications (CPU, RAM, disk), and network configuration data including all IPv4 addresses and network adapter MAC addresses.

```
Select CIP-B101 System Info Collection
C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\scripts>sys_info.bat
Please wait... Checking system information.
Collection complete. See outputs\sys_info_report.txt
Press any key to continue . . .

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\scripts>type ..\outputs\sys_info_report.txt
=====
WINDOWS INFO
=====
OS Name: Microsoft Windows 10 Pro
OS Version: 10.0.19045 N/A Build 19045
BIOS Version: Dell Inc. 1.9.3, 3/8/2018
System Type: x64-based PC
=====
HARDWARE INFO
=====
Total Physical Memory: 16,026 MB
Name
Intel(R) Core(TM) i7-7820HQ CPU @ 2.90GHz
Model SAMSUNG MZVLB256HAHQ-000L7 Name Size
\\.\PHYSICALDRIVE0 256052966400
=====
NETWORK INFO
=====
IPv4 Address. . . . . : 172.16.0.2
IPv4 Address. . . . . : 172.25.240.1
IPv4 Address. . . . . : 172.19.112.1
IPv4 Address. . . . . : 192.168.56.1
IPv4 Address. . . . . : 192.168.102.1
IPv4 Address. . . . . : 192.168.199.1
IPv4 Address. . . . . : 192.168.18.4
Description MACAddress
Microsoft Kernel Debug Network Adapter
Intel(R) Dual Band Wireless-AC 8265 30:24:32:B5:1B:CF
Intel(R) Ethernet Connection (5) I219-LM 10:65:30:83:F8:B5
Microsoft Wi-Fi Direct Virtual Adapter 30:24:32:B5:1B:D0
WAN Miniport (SSTP)
WAN Miniport (IKEv2)
WAN Miniport (L2TP)
WAN Miniport (PPTP)
WAN Miniport (PPPOE)
WAN Miniport (IP) 76:B6:20:52:41:53
WAN Miniport (IPv6) 90:C7:20:52:41:53
WAN Miniport (Network Monitor) A0:BD:20:52:41:53
Microsoft Wi-Fi Direct Virtual Adapter 32:24:32:B5:1B:CF
Bluetooth Device (Personal Area Network) 30:24:32:B5:1B:03
VirtualBox Host-Only Ethernet Adapter 0A:00:27:00:00:0A
VMware Virtual Ethernet Adapter for VMnet1 00:50:56:C0:00:01
```

Screenshot 11: Executing sys_info.bat and viewing the generated sys_info_report.txt

Table I.1: Essential Batch File Commands

Command	Description	Forensic Application
@ECHO OFF	Suppress command echoing	Cleaner script output
ECHO message	Display text to console	Status messages during collection
ECHO. > file	Write to file (create/overwrite)	Initialize report files
ECHO text >> file	Append text to file	Build multi-section reports
command >> file	Append command output to file	Collect command results
PAUSE	Wait for keypress	Review output before continuing
REM comment	Add comments to script	Document script purpose

Part J: PATH Modification and Global Script Execution

Modifying the PATH environment variable allows executables and scripts to be run from any directory on the system without specifying their full path. The **setx** command permanently modifies environment variables for future sessions (unlike **set**, which only modifies the current session). In this exercise, the workspace4 directory was added to the system PATH using setx, enabling the sys_info.bat script to be executed from any location on the system, including the root directory C:\. Understanding PATH modification is forensically relevant because malicious actors sometimes add their own directories to the PATH to ensure their tools are executed preferentially, a technique known as DLL search order hijacking or executable hijacking.

The PATH modification was performed using the command `setx PATH "%PATH%;C:/Users/Public\CIP-B101_Lab1C_Almustapha_Yusuf\workspace4"`. The system responded with a warning that the data was truncated to 1024 characters, which is a known limitation of the setx command for environment variables exceeding this length. Despite this warning, the modification was successful. When sys_info.bat was subsequently executed from the root directory (C:\), the script was found via PATH and began execution. However, the output redirections using relative paths (outputs\sys_info_report.txt) failed because the relative path does not resolve correctly from C:\. This demonstrates an important principle: while PATH enables locating executables globally, scripts that use relative internal paths must account for their working directory. The full sys_info_report.txt was still accessible via its absolute path, confirming the report data was intact.

J.1 PATH Modification and Cross-Directory Execution

```
CIP-B101 System Info Collection

C:\>cd \Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf>copy scripts\sys_info.bat workspace4\sys_info.bat
Overwrite workspace4\sys_info.bat? (Yes/No/All): Y
1 file(s) copied.

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf>dir workspace4
Volume in drive C has no label.
Volume Serial Number is E2F9-902F

Directory of C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\workspace4

08/09/2026 10:57 PM <DIR> .
08/09/2026 10:57 PM <DIR> ..
08/09/2026 10:52 PM 1,393 sys_info.bat
1 File(s) 1,393 bytes
2 Dir(s) 5,573,087,232 bytes free

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf>setx PATH "%PATH%;C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\workspace4"
WARNING: The data being saved is truncated to 1024 characters.
SUCCESS: Specified value was saved.

C:\Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf>cd \

C:\>sys_info.bat
Please wait... Checking system information.
The system cannot find the path specified.
The system cannot find the path specified.
The system cannot find the path specified.
The system cannot find the path specified.

C:\>type \Users\Public\CIP-B101_Lab1C_Almustapha_Yusuf\outputs\sys_info_report.txt
=====
WINDOWS INFO
=====
OS Name: Microsoft Windows 10 Pro
OS Version: 10.0.19045 N/A Build 19045
BIOS Version: Dell Inc. 1.9.3, 3/8/2018
System Type: x64-based PC
=====
HARDWARE INFO
=====
Total Physical Memory: 16,026 MB
Name
Intel(R) Core(TM) i7-7820HQ CPU @ 2.90GHz
Model Name Size
SAMSUNG MZVLB256HAHQ-000L7 \\.\PHYSICALDRIVE0 256052966400
=====
NETWORK INFO
=====
IPv4 Address. . . . . : 172.16.0.2
IPv4 Address. . . . . : 172.25.240.1
IPv4 Address. . . . . : 172.19.112.1
IPv4 Address. . . . . : 192.168.56.1
IPv4 Address. . . . . : 192.168.102.1
IPv4 Address. . . . . : 192.168.199.1
IPv4 Address. . . . . : 192.168.18.4
Description MACAddress
Microsoft Kernel Debug Network Adapter
Intel(R) Dual Band Wireless-AC 8265 30:24:32:B5:1B:CF
Intel(R) Ethernet Connection (5) I219-LM 10:65:30:83:F8:B5
Microsoft Wi-Fi Direct Virtual Adapter 30:24:32:B5:1B:D0
WAN Miniport (802.11)
WAN Miniport (IEEE802.3)
WAN Miniport (L2TP)
WAN Miniport (PPTP)
WAN Miniport (PPPOE)
WAN Miniport (IP) 76:86:20:52:41:53
WAN Miniport (IPv6) 90:C7:20:52:41:53
WAN Miniport (Network Monitor) A0:BD:20:52:41:53
Microsoft Wi-Fi Direct Virtual Adapter 32:24:32:B5:1B:CF
Bluetooth Device (Personal Area Network) 30:24:32:B5:1B:D3
VirtualBox Host-Only Ethernet Adapter 0A:00:27:00:00:0A
VMware Virtual Ethernet Adapter for VMnet1 00:50:56:C0:00:01
Hyper-V Virtual Switch Extension Adapter 00:50:56:C0:00:08
VMware Virtual Ethernet Adapter for VMnet8 00:50:56:C0:00:08
Hyper-V Virtual Ethernet Adapter 00:15:5D:24:A0:C8
Hyper-V Virtual Switch Extension Adapter 00:15:5D:D6:EC:05
WinTun Userspace Tunnel
```

Screenshot 12.0: Copying sys_info.bat to workspace4, modifying PATH with setx, and executing from root

```
CIP-B101 System Info Collection
=====
WINDOWS INFO
=====
OS Name: Microsoft Windows 10 Pro
OS Version: 10.0.19045 N/A Build 19045
BIOS Version: Dell Inc. 1.9.3, 3/8/2018
System Type: x64-based PC
=====
HARDWARE INFO
=====
Total Physical Memory: 16,026 MB
Name
Intel(R) Core(TM) i7-7820HQ CPU @ 2.90GHz
Model Name Size
SAMSUNG MZVLB256HAHQ-000L7 \\.\PHYSICALDRIVE0 256052966400
=====
NETWORK INFO
=====
IPv4 Address. . . . . : 172.16.0.2
IPv4 Address. . . . . : 172.25.240.1
IPv4 Address. . . . . : 172.19.112.1
IPv4 Address. . . . . : 192.168.56.1
IPv4 Address. . . . . : 192.168.102.1
IPv4 Address. . . . . : 192.168.199.1
IPv4 Address. . . . . : 192.168.18.4
Description MACAddress
Microsoft Kernel Debug Network Adapter
Intel(R) Dual Band Wireless-AC 8265 30:24:32:B5:1B:CF
Intel(R) Ethernet Connection (5) I219-LM 10:65:30:83:F8:B5
Microsoft Wi-Fi Direct Virtual Adapter 30:24:32:B5:1B:D0
WAN Miniport (802.11)
WAN Miniport (IEEE802.3)
WAN Miniport (L2TP)
WAN Miniport (PPTP)
WAN Miniport (PPPOE)
WAN Miniport (IP) 76:86:20:52:41:53
WAN Miniport (IPv6) 90:C7:20:52:41:53
WAN Miniport (Network Monitor) A0:BD:20:52:41:53
Microsoft Wi-Fi Direct Virtual Adapter 32:24:32:B5:1B:CF
Bluetooth Device (Personal Area Network) 30:24:32:B5:1B:D3
VirtualBox Host-Only Ethernet Adapter 0A:00:27:00:00:0A
VMware Virtual Ethernet Adapter for VMnet1 00:50:56:C0:00:01
Hyper-V Virtual Switch Extension Adapter 00:50:56:C0:00:08
VMware Virtual Ethernet Adapter for VMnet8 00:50:56:C0:00:08
Hyper-V Virtual Ethernet Adapter 00:15:5D:24:A0:C8
Hyper-V Virtual Switch Extension Adapter 00:15:5D:D6:EC:05
WinTun Userspace Tunnel
```

Screenshot 12.1: Full sys_info_report.txt content view showing complete system information

Table J.1: set vs setx Comparison

Feature	set	setx
Scope	Current CMD session only	Persistent (future sessions)
Persistence	Lost when CMD closes	Survives system restart
Max Length	8191 characters	1024 characters
Requires Admin	No	No (for user vars)
Syntax	set VAR=value	setx VAR "value"
Forensic Use	Test variables safely	Detect malicious PATH changes

Reflection

This laboratory exercise provided comprehensive hands-on experience with Windows Command Prompt operations that are directly applicable to digital forensic investigations. Through systematic exploration of navigation commands, file operations, system information gathering, network analysis, text searching, permission auditing, process monitoring, batch scripting, and PATH manipulation, I developed a thorough understanding of how Windows CMD serves as a powerful forensic tool. The most valuable insight gained was understanding how each command relates to specific forensic workflows: directory navigation for evidence location, file operations for evidence preservation, systeminfo and wmic for establishing the baseline environment, ipconfig and network commands for communications analysis, findstr for keyword searching across evidence sets, icacls for access control auditing, and batch files for automating repetitive collection tasks.

The batch file scripting exercise was particularly instructive, demonstrating how automation can standardize evidence collection procedures and ensure consistency across multiple examinations. The PATH modification exercise highlighted an important security consideration: the ability to add directories to PATH can be exploited by threat actors to execute malicious code, making PATH auditing a necessary step in forensic investigations. Additionally, the comparison between set (temporary) and setx (persistent) environment variable modification underscored the difference between transient and persistent system changes, a distinction that is critical when analyzing the timeline of a security incident. Overall, this lab reinforced the importance of command-line proficiency in digital forensics and provided practical skills that will be essential in professional forensic examination scenarios.